

MODEL QUESTION PAPER
Refrigeration & Air conditioning

Time :3Hours

Max. Marks :75

PART A

I Choose the correct answer

1	Mechatronics engineering is a branch of engineering that involves the study of -----, ----- systems and information technology.	M1.01	U
2	A ----- is a device that produces an output signal for the purpose of sensing a physical phenomenon.	M1.03	U
3	A -----chain is a model for a mechanical system composed of rigid bodies linked together by joints.	M2.01	U
4	----- is a brushless DC motor that rotates in steps.	M2.07	U
5	An ----- converts such an input signal into the required form of mechanical energy.	M2.05	U
6	A -----is a digital computer that automates industrial processes and monitors inputs/outputs.	M3.01	U
7	PLC programming software offers programming in ----- Logic, or “C”.	M3.02	U
8	-----is the process of using technology to complete tasks otherwise performed by humans.	M4.01	U
9	----- control systems are essential for robotics, as they allow robots to adjust their actions based on the desired outcomes and the actual outcomes.	M4.03	U

(9x1=9 marks)

PART B

II Answer any eight questions from the following .Each question carries 3 marks

1	List the applications of mechatronics.	M1.01	U
2	Draw force sensor with simple diagram and mark the parts.	M1.03	U
3	Explain geneva mechanism.	M2.01	U
4	Explain pressure control valve.	M2.03	U
5	Define solid state switch.	M2.06	U
6	Explain ladder diagram.	M3.02	U
7	Explain shift registers.	M3.04	U
8	Explain data handling.	M3.05	U
9	List the classification of robotics.	M4.01	U
10	Explain robot activation methods.	M4.03	U

(8x3=24 marks)

PART C

III Answer ALL questions. Each question carries 7 marks

III	Explain measurement system terminology.	M1.02	U
OR			
IV	Explain the working displacement and force sensor with a neat sketch.	M1.03	U
V	Explain the factors considering during the selection of sensors.	M1.04	U
OR			
VI	Explain the working of automatic water level controller with a neat sketch.	M1.05	U
VII	Describe the working of hydraulic system with a neat sketch.	M2.02	U
OR			
VIII	Explain cylinder sequencing with suitable diagram.	M2.04	U
IX	Explain the working of mechanical switches with suitable sketch.	M2.06	U
OR			
X	Explain the Working principle of Stepper motors.	M2.07	U
XI	Explain programmable logic controller with a suitable sketch.	M3.01	U
OR			
XII	Explain PLC mnemonics.	M3.03	U
XIII	Explain selection of PLC.	M3.05	A
OR			
XIV	Explain bathroom scale.	M3.06	U
XV	Explain robotic applications in manufacturing.	M4.02	U
OR			
XVI	Explain assembly and inspection in robotics.	M4.03	U
XVII	Explain programming of robot.	M4.04	A
OR			
XVIII	Explain pick and place robot	M4.05	U

(6x7=42 marks)